

Uniform rates

Due to clever tax evasions, the Indian Government may force telecom companies to use uniform revenue models. This covers all the services offered, including mobile connections and Internet services.

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ISPs – The Fine Lines

Your ISP's plans might not be as straightforward as they seem...

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VSNL first introduced internet services in India in 1995. One should think that when the internet is entering its fifteenth year in Indian history (starting August 2009), our access to the network of networks would be a piece of cake. Then how come we end up with mud on our faces? Perhaps the reason is we don't know enough about the chaps and chips that provide us this access. Perhaps we need to know our internet service providers a little better.

Internet service providers

Any organisation that provides you access to the internet could be termed an internet service provider (ISP) – sometimes also called an internet access provider (IAP). Your home PC could be linked to the internet using a phone line modem, digital subscriber line (DSL) or a cable modem, which serves as an interface between your machine and the machine of your ISP – a 'server'. There are more than 150 ISPs in India with some popular ones being BSNL, Airtel, Reliance and Tata.

Types of connections

An ISP would typically present you with a myriad of options and you'll have to choose how you want to connect to the internet. These are basically either dialup connections or broadband ones.

Dialup

A dial-up account (or analogue connection) simply means using your telephone line to access the internet. This is slow and keeps your phone line occupied, implying that you can't talk on the phone and browse simultaneously. Downloading audio/video streams could be troublesome while using dial-ups. Besides, charges are levied over and above the regular charges that you would have to pay for a telephone call of the same duration. So if you've browsed for an hour, they would, likely, charge you for the cost of the telephone call (that's lasted for an hour) ~plus~ the cost of internet usage.

Sometimes, ISPs advertise "high-speed dialup connections". These may offer to provide speeds as high as 256 kbps (the bottom-line for broadband service speeds in India) – but what you get in reality is far from the typically seamless broadband experience. What ISPs actually do here is use a "handshake" – a shortened logon process and a different data-compression technique to speed up the process. Besides, they use an 'Acceleration Server' – a small-business Microsoft server that lets the ISP connect to the web through broadband. However, in the end, it's the telephone line that will receive the data and there's only so much data it can sustain at a particular moment – irrespective of what, and how much, is fed from the other end. Typically, dialups promise a maximum speed of 56 kbps, which is pretty much equivalent to snail mail by present day standards. A dialup connection is, therefore, the last resort of a frantic webophile. All the rest use broadband.

Broadband

According to the definition by the Department of Telecommunication Broadband Policy published in 2004, 'broadband' refers to an 'always-on' data connection that is able to support interactive services including internet access and has the capability of the minimum download speed of 256 kilobits per second (256 kbps) to an individual subscriber from the point of presence (POP). The major access technologies for broadband are:

- **DSL:** Digital subscriber line is a technology that provides access using a dedicated digital circuit from your residence (or office) to the ISP's central office. It allows high-speed data exchange over existing telephone lines, usually using a modem at each end. DSL connection speeds range from 128 Kbps to 8 Mbps. The DSL bouquet of technologies includes ADSL, SDSL and VDSL.
- **ADSL:** The asymmetric digital subscriber line technology is quite commonly used in India and is just another kind of DSL. The only difference here is that it provides more bandwidth downstream than upstream, so uploaders may sometimes feel that they've got a raw deal.
- **SDSL:** The Symmetric Digital Subscriber Line is another DSL connection, with a minor difference: the technology allows the same data rates for upstream as well as downstream traffic. The same telephone wires, however, cannot be simultaneously used for voice connections.
- **VDSL:** The 'V' stands for 'Very High' and VDSL technology, therefore assures you of super-fast data rates. However, it is usually meant for relatively short distances – the data rate being in direct proportion to the distance between the subscriber and the server.

Cable

Cable TV has been around a lot longer than the internet, with coaxial wires linking millions of Indians with cable operators. Cable ISPs provide internet access through existing lines.

Satellite internet service

Geo-stationary satellites allow users to access the web using the IoS (Internet over Satellite) technology.

Wireless Internet Access

'Wireless' is a magic word for techies – and technologies like Wi-Fi, WAP (Wireless Access Protocol), CDMA and GSM, offer access to the internet by making wires disappear, replacing the wires with radio waves.



TATA plans to enter the ISP field with an aggressive strategy • CERT India has joined forces with Microsoft